

Semi-Supervised Deep Sobolev Regression: Estimation, Variable Selection, and Beyond

Chenguang Duan

Wuhan University

Joint work with Zhao Ding, Yuling Jiao, and Jerry Zhijian Yang

The 3rd Joint Conference on Statistics and Data Science in China
Hangzhou, June 30 to July 4, 2025



武漢大學
WUHAN UNIVERSITY

- ① Introduction and Background
- ② Sobolev-Penalized Regression
- ③ Semi-Supervised Deep Sobolev Regression
- ④ Applications and Numerical Experiments
 - Derivative Estimations
 - Nonparametric Variable Selection
 - Inverse Problems
- ⑤ Conclusions

Nonparametric regression model

$$Y_i = f_0(X_i) + \xi_i$$

- ▶ $X_1, \dots, X_n \sim \text{i.i.d. } \mu$.
- ▶ $\xi_1, \dots, \xi_n \sim \text{i.i.d. } N(0, \sigma^2 I_d)$.
- ▶ $\xi_{1:n}$ is independent of $X_{1:n}$.

The goal of the regression is to find the unknown function f_0 using labeled data $\{(X_i, Y_i)\}_{i=1}^n$.

Nonparametric regression model

$$Y_i = f_0(X_i) + \xi_i$$

- ▶ $X_1, \dots, X_n \sim \text{i.i.d. } \mu$.
- ▶ $\xi_1, \dots, \xi_n \sim \text{i.i.d. } N(0, \sigma^2 I_d)$.
- ▶ $\xi_{1:n}$ is independent of $X_{1:n}$.

The goal of the regression is to find the unknown function f_0 using labeled data $\{(X_i, Y_i)\}_{i=1}^n$.

Least-squares regression

$$f_0 = \arg \min_{f \text{ measurable}} L(f) := \mathbb{E}_{(X,Y)} [(Y - f(X))^2]$$

Nonparametric regression model

$$Y_i = f_0(X_i) + \xi_i$$

- ▶ $X_1, \dots, X_n \sim \text{i.i.d. } \mu$.
- ▶ $\xi_1, \dots, \xi_n \sim \text{i.i.d. } N(0, \sigma^2 I_d)$.
- ▶ $\xi_{1:n}$ is independent of $X_{1:n}$.

The goal of the regression is to find the unknown function f_0 using labeled data $\{(X_i, Y_i)\}_{i=1}^n$.

Least-squares regression

$$f_0 = \arg \min_{f \text{ measurable}} L(f) := \mathbb{E}_{(X,Y)} [(Y - f(X))^2]$$

- ▶ The expectation is intractable.
- ▶ Minimization over the measurable function class is intractable.

Nonparametric regression model

$$Y_i = f_0(X_i) + \xi_i$$

- ▶ $X_1, \dots, X_n \sim \text{i.i.d. } \mu$.
- ▶ $\xi_1, \dots, \xi_n \sim \text{i.i.d. } N(0, \sigma^2 I_d)$.
- ▶ $\xi_{1:n}$ is independent of $X_{1:n}$.

The goal of the regression is to find the unknown function f_0 using labeled data $\{(X_i, Y_i)\}_{i=1}^n$.

Least-squares regression

$$f_0 = \arg \min_{f \text{ measurable}} L(f) := \mathbb{E}_{(X,Y)} [(Y - f(X))^2]$$

- ▶ The expectation is intractable.
- ▶ Minimization over the measurable function class is intractable.

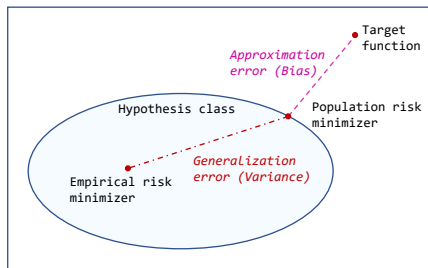
Monte Carlo Approximation
Parameterization

$$\hat{f}_n \in \arg \min_{f \in \mathcal{F}} \hat{L}_n(f) := \frac{1}{n} \sum_{i=1}^n (Y_i - f(X_i))^2$$

$$\hat{f}_n \in \arg \min_{f \in \mathcal{F}} \hat{L}_n(f) := \frac{1}{n} \sum_{i=1}^n (Y_i - f(X_i))^2$$

Error decomposition

$$\mathbb{E}_S [\|\hat{f}_n - f_0\|_2^2] \leq \underbrace{\inf_{f \in \mathcal{F}} \|f - f_0\|_{L^2(\Omega)}}_{\text{Approximation error}} + \underbrace{\mathbb{E}_S \left[\sup_{f \in \mathcal{F}} (L(f) - \hat{L}_n(f)) \right]}_{\text{Generalization error}}$$



Bias-Variance trade-off

Choose the hypothesis class as a **deep neural network class**.

A neural network $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$ is a function defined by

$$f(x) = T_L(\rho(T_{L-1}(\cdots \rho(T_0(x)) \cdots))), \quad (\text{DNN})$$

where the activation function ρ is applied component-wisely and $T_\ell(x) = A_\ell x + b_\ell$ is an affine transformation with $A_\ell \in \mathbb{R}^{N_{\ell+1} \times N_\ell}$ and $b_\ell \in \mathbb{R}^{N_\ell}$ for $\ell = 0, \dots, L$. Then the ρ -activated deep neural network class $N_\rho(L, S, B)$ is defined as

$$N_\rho(L, S, B) = \left\{ f(x) \text{ has the form of (DNN)} : \sum_{0 \leq \ell \leq L+1} (\|A_\ell\|_0 + \|b_\ell\|_2) \leq S, \|f\|_{L^\infty(\mathbb{R}^d)} \leq B \right\},$$

where L is called the depth of neural networks, S represents the number of non-zero parameters, and B is the uniform bound of neural networks.

Approximation error of DNN

- ▶ Approximation in L^p -norm
(Dmitry Yarotsky 2017, Zuowei Shen et al. 2019, Johannes Schmidt-Hieber 2020, Jianfeng Lu et al. 2020, Yuling Jiao et al. 2023)
- ▶ Approximation in Sobolev norms
(Ingo Gühring et al. 2021, **Chenguang Duan et al. 2022**)
- ▶ Approximation with Lipschitz constraint
(Jian Huang et al. 2022, Yuling Jiao et al. 2023, **Zhao Ding et al. 2024**)

Generalization error of DNN

Empirical process + Sample complexity of DNN

- ▶ (Local) Rademacher complexity (Peter L. Bartlett et al. 2002, 2005)
- ▶ Size-independent Rademacher complexity of DNN (Noah Golowich et al. 2018)
- ▶ VC-dimension bound for DNN (Martin Anthony et al. 1999, Peter L. Bartlett et al. 1998, 2019)

Suppose that $f_0 \in \mathcal{H}^s(\Omega)$, then the minimax optimal rate of an estimator $\hat{f}_n$ is given as:

$$\mathbb{E}_S \left[\|\hat{f}_n - f_0\|_{L^2(\Omega)}^2 \right] = \mathcal{O} \left(n^{-\frac{2s}{d+2s}} \right).$$

Suppose that $f_0 \in \mathcal{H}^s(\Omega)$, then the minimax optimal rate of an estimator $\hat{f}_n$ is given as:

$$\mathbb{E}_S \left[\|\hat{f}_n - f_0\|_{L^2(\Omega)}^2 \right] = \mathcal{O} \left(n^{-\frac{2s}{d+2s}} \right).$$

The convergence in L^2 -norm can **NOT** imply the convergence of gradient.

► For example,

$$\hat{f}_n(x) = \frac{\sin(nx) \log n}{n} \xrightarrow{L^p} 0, \quad \text{for each } 1 \leq p \leq \infty.$$

However, $\hat{f}'_n(x) = \cos(nx) \log n \rightarrow +\infty$.

Suppose that $f_0 \in \mathcal{H}^s(\Omega)$, then the minimax optimal rate of an estimator $\hat{f}_n$ is given as:

$$\mathbb{E}_S \left[\|\hat{f}_n - f_0\|_{L^2(\Omega)}^2 \right] = \mathcal{O} \left(n^{-\frac{2s}{d+2s}} \right).$$

The convergence in L^2 -norm can **NOT** imply the convergence of gradient.

► For example,

$$\hat{f}_n(x) = \frac{\sin(nx) \log n}{n} \xrightarrow{L^p} 0, \quad \text{for each } 1 \leq p \leq \infty.$$

However, $\hat{f}'_n(x) = \cos(nx) \log n \rightarrow +\infty$.

How can we simultaneously estimate both the regression function f_0 and its gradient ∇f_0 ?

Applications in inverse problems, nonparametric variable selection, generative learning ...

- ① Introduction and Background
- ② Sobolev-Penalized Regression
- ③ Semi-Supervised Deep Sobolev Regression
- ④ Applications and Numerical Experiments
 - Derivative Estimations
 - Nonparametric Variable Selection
 - Inverse Problems
- ⑤ Conclusions

Least-squares regression with gradient penalty

$$L^\lambda(f) := \underbrace{\mathbb{E}_{(X,Y)} [(Y - f(X))^2]}_{\text{least-squares}} + \underbrace{\lambda |f|_{H^1(\Omega)}^2}_{\text{gradient penalty}} = \|f - f_0\|_{L^2(\Omega)}^2 + \sigma^2 + \lambda |f|_{H^1(\Omega)}.$$

Least-squares regression with gradient penalty

$$L^\lambda(f) := \underbrace{\mathbb{E}_{(X,Y)} [(Y - f(X))^2]}_{\text{least-squares}} + \underbrace{\lambda |f|_{H^1(\Omega)}^2}_{\text{gradient penalty}} = \|f - f_0\|_{L^2(\Omega)}^2 + \sigma^2 + \lambda |f|_{H^1(\Omega)}.$$

Variational problem

Find $f^\lambda \in H^1(\Omega)$, such that $\delta L^\lambda(f^\lambda, v) = 0$ for each $v \in H^1(\Omega)$, that is,

$$(f^\lambda - f_0, v)_{L^2(\Omega)} + \lambda (\nabla(f^\lambda - f_0), \nabla v)_{L^2(\Omega)} = \lambda (\nabla f_0, \nabla v)_{L^2(\Omega)}.$$

Least-squares regression with gradient penalty

$$L^\lambda(f) := \underbrace{\mathbb{E}_{(X,Y)} [(Y - f(X))^2]}_{\text{least-squares}} + \underbrace{\lambda |f|_{H^1(\Omega)}^2}_{\text{gradient penalty}} = \|f - f_0\|_{L^2(\Omega)}^2 + \sigma^2 + \lambda |f|_{H^1(\Omega)}.$$

Variational problem

Find $f^\lambda \in H^1(\Omega)$, such that $\delta L^\lambda(f^\lambda, v) = 0$ for each $v \in H^1(\Omega)$, that is,

$$(f^\lambda - f_0, v)_{L^2(\Omega)} + \lambda (\nabla(f^\lambda - f_0), \nabla v)_{L^2(\Omega)} = \lambda (\nabla f_0, \nabla v)_{L^2(\Omega)}.$$

Notice that f^λ is the solution to the elliptic equation

$$\begin{cases} -\lambda \Delta f + f = f_0, & \text{in } \Omega, \\ \nabla f \cdot \mathbf{n} = 0, & \text{on } \partial\Omega. \end{cases}$$

Substituting $v = f^\lambda - f_0$ into $\delta L^\lambda(f^\lambda, v) = 0$ implies

$$\begin{aligned} & \overbrace{\|f^\lambda - f_0\|_{L^2(\Omega)}^2}^{\text{interior } L^2 \text{ error}} + \lambda \overbrace{|f^\lambda - f_0|_{H^1(\Omega)}^2}^{\text{interior gradient error}} \\ &= \lambda(\Delta f_0, f_0 - f^\lambda)_{L^2(\Omega)} \leq \lambda \|\Delta f_0\|_{L^2(\Omega)} \|f^\lambda - f_0\|_{L^2(\Omega)}. \end{aligned}$$

Substituting $v = f^\lambda - f_0$ into $\delta L^\lambda(f^\lambda, v) = 0$ implies

$$\begin{aligned} & \overbrace{\|f^\lambda - f_0\|_{L^2(\Omega)}^2}^{\text{interior } L^2 \text{ error}} + \lambda \overbrace{\|f^\lambda - f_0\|_{H^1(\Omega)}^2}^{\text{interior gradient error}} \\ &= \lambda(\Delta f_0, f_0 - f^\lambda)_{L^2(\Omega)} \leq \lambda \|\Delta f_0\|_{L^2(\Omega)} \|f^\lambda - f_0\|_{L^2(\Omega)}. \end{aligned}$$

► Interior L^2 error:

$$\|f^\lambda - f_0\|_{L^2(\Omega)}^2 \leq \lambda^2 \|\Delta f_0\|_{L^2(\Omega)}^2.$$

Substituting $v = f^\lambda - f_0$ into $\delta L^\lambda(f^\lambda, v) = 0$ implies

$$\begin{aligned} & \overbrace{\|f^\lambda - f_0\|_{L^2(\Omega)}^2}^{\text{interior } L^2 \text{ error}} + \lambda \overbrace{|f^\lambda - f_0|_{H^1(\Omega)}^2}^{\text{interior gradient error}} \\ &= \lambda(\Delta f_0, f_0 - f^\lambda)_{L^2(\Omega)} \leq \lambda \|\Delta f_0\|_{L^2(\Omega)} \|f^\lambda - f_0\|_{L^2(\Omega)}. \end{aligned}$$

► Interior L^2 error:

$$\|f^\lambda - f_0\|_{L^2(\Omega)}^2 \leq \lambda^2 \|\Delta f_0\|_{L^2(\Omega)}^2.$$

► Interior gradient error:

$$|f^\lambda - f_0|_{H^1(\Omega)}^2 \leq \lambda \|\Delta f_0\|_{L^2(\Omega)}^2.$$

Substituting $v = f^\lambda - f_0$ into $\delta L^\lambda(f^\lambda, v) = 0$ implies

$$\begin{aligned} & \overbrace{\|f^\lambda - f_0\|_{L^2(\Omega)}^2}^{\text{interior } L^2 \text{ error}} + \lambda \overbrace{|f^\lambda - f_0|_{H^1(\Omega)}^2}^{\text{interior gradient error}} \\ &= \lambda(\Delta f_0, f_0 - f^\lambda)_{L^2(\Omega)} \leq \lambda \|\Delta f_0\|_{L^2(\Omega)} \|f^\lambda - f_0\|_{L^2(\Omega)}. \end{aligned}$$

► Interior L^2 error:

$$\|f^\lambda - f_0\|_{L^2(\Omega)}^2 \leq \lambda^2 \|\Delta f_0\|_{L^2(\Omega)}^2.$$

► Interior gradient error:

$$|f^\lambda - f_0|_{H^1(\Omega)}^2 \leq \lambda \|\Delta f_0\|_{L^2(\Omega)}^2.$$

The Sobolev-penalized regressor f^λ is an estimator of f_0 in both L^2 -norm and H^1 -semi-norm.

- 1 Introduction and Background
- 2 Sobolev-Penalized Regression
- 3 Semi-Supervised Deep Sobolev Regression**
- 4 Applications and Numerical Experiments
 - Derivative Estimations
 - Nonparametric Variable Selection
 - Inverse Problems
- 5 Conclusions

Sobolev-Penalized Risk

$$L^\lambda(f) := \mathbb{E}_{(X,Y)} [(Y - f(X))^2] + \lambda |f|_{H^1(\Omega)}^2$$

- ▶ The expectation is intractable.
- ▶ Minimization over the measurable function class is intractable.

Monte Carlo Approximation

Parameterization

Sobolev-Penalized Risk

$$L^\lambda(f) := \mathbb{E}_{(X,Y)} [(Y - f(X))^2] + \lambda |f|_{H^1(\Omega)}^2$$

- ▶ The expectation is intractable.
- ▶ Minimization over the measurable function class is intractable.

Monte Carlo Approximation

Parameterization

Empirical Sobolev-Penalized Risk Minimization

$$\hat{f}_{n,m}^\lambda \in \arg \min_{f \in \text{conv}(\mathcal{F})} \hat{L}_{n,m}^\lambda(f) = \frac{1}{n} \sum_{i=1}^n (Y_i - f(X_i))^2 + \frac{\lambda}{m} \sum_{j=1}^m \|\nabla f(Z_j)\|_2^2.$$

- ▶ Labeled data: $(X_i, Y_i) \sim \text{i.i.d. } P$
- ▶ Unlabeled data: $Z_j \sim \text{i.i.d. } \text{unif}(\Omega)$

Sobolev-Penalized Risk

$$L^\lambda(f) := \mathbb{E}_{(X,Y)} [(Y - f(X))^2] + \lambda |f|_{H^1(\Omega)}^2$$

- ▶ The expectation is intractable.
- ▶ Minimization over the measurable function class is intractable.

Monte Carlo Approximation

Parameterization

Empirical Sobolev-Penalized Risk Minimization

$$\hat{f}_{n,m}^\lambda \in \arg \min_{f \in \text{conv}(\mathcal{F})} \hat{L}_{n,m}^\lambda(f) = \frac{1}{n} \sum_{i=1}^n (Y_i - f(X_i))^2 + \frac{\lambda}{m} \sum_{j=1}^m \|\nabla f(Z_j)\|_2^2.$$

- ▶ Labeled data: $(X_i, Y_i) \sim \text{i.i.d. } P$
- ▶ Unlabeled data: $Z_j \sim \text{i.i.d. } \text{unif}(\Omega)$

Semi-supervised framework

- **A1. Sub-Gaussian noise.** The noise ξ is sub-Gaussian with mean 0 and finite variance proxy σ^2 .

- ▶ **A1. Sub-Gaussian noise.** The noise ξ is sub-Gaussian with mean 0 and finite variance proxy σ^2 .
- ▶ **A2. Bounded hypothesis.** There exists an absolute positive constant B_0 , such that $\sup_{x \in \Omega} |f_0(x)| \leq B_0$. Further, functions in hypothesis class $\mathcal{F}$ are also bounded, that is, $\sup_{x \in \Omega} |f(x)| \leq B_0$.

- ▶ **A1. Sub-Gaussian noise.** The noise ξ is sub-Gaussian with mean 0 and finite variance proxy σ^2 .
- ▶ **A2. Bounded hypothesis.** There exists an absolute positive constant B_0 , such that $\sup_{x \in \Omega} |f_0(x)| \leq B_0$. Further, functions in hypothesis class $\mathcal{F}$ are also bounded, that is, $\sup_{x \in \Omega} |f(x)| \leq B_0$.
- ▶ **A3. Bounded derivatives of hypothesis.** There exists positive constants $\{B_{1,k}\}_{k=1}^d$, such that $\sup_{x \in \Omega} |D_k f_0(x)| \leq B_{1,k}$ for $1 \leq k \leq d$. Further, the first-order partial derivatives of functions in hypothesis class $\mathcal{F}$ are also bounded, i.e., $\sup_{x \in \Omega} |D_k f(x)| \leq B_{1,k}$ for each $1 \leq k \leq d$ and $f \in \mathcal{F}$. Denote by $B_1^2 := \sum_{k=1}^d B_{1,k}^2$.

- ▶ **A1. Sub-Gaussian noise.** The noise ξ is sub-Gaussian with mean 0 and finite variance proxy σ^2 .
- ▶ **A2. Bounded hypothesis.** There exists an absolute positive constant B_0 , such that $\sup_{x \in \Omega} |f_0(x)| \leq B_0$. Further, functions in hypothesis class $\mathcal{F}$ are also bounded, that is, $\sup_{x \in \Omega} |f(x)| \leq B_0$.
- ▶ **A3. Bounded derivatives of hypothesis.** There exists positive constants $\{B_{1,k}\}_{k=1}^d$, such that $\sup_{x \in \Omega} |D_k f_0(x)| \leq B_{1,k}$ for $1 \leq k \leq d$. Further, the first-order partial derivatives of functions in hypothesis class $\mathcal{F}$ are also bounded, i.e., $\sup_{x \in \Omega} |D_k f(x)| \leq B_{1,k}$ for each $1 \leq k \leq d$ and $f \in \mathcal{F}$. Denote by $B_1^2 := \sum_{k=1}^d B_{1,k}^2$.
- ▶ **A4. Regularity of regression function.** The regression function satisfies $\Delta f_0 \in L^2(\Omega)$ and $\nabla f_0 \cdot \mathbf{n} = 0$ a.e. on $\partial\Omega$, where $\mathbf{n}$ is the unit normal to the boundary.

Oracle inequality

Under **A1** to **A4**. Suppose $n \geq \log N(B_0\delta, \mathcal{F}, L^2(\mathcal{D}))$ and $m \geq \max_k \log N(B_{1,k}\delta, D_k\mathcal{F}, L^2(\mathcal{S}))$. Then

$$\mathbb{E} [\|\hat{f}_{n,m}^\lambda - f_0\|_{L^2(\Omega)}^2] \lesssim \beta\lambda^2 + \varepsilon_{\text{app}}(\mathcal{F}, \lambda) + \varepsilon_{\text{gen}}(\mathcal{F}, n) + \varepsilon_{\text{gen}}^{\text{reg}}(\nabla\mathcal{F}, m),$$

$$\mathbb{E} [\|\nabla(\hat{f}_{n,m}^\lambda - f_0)\|_{L^2(\Omega)}^2] \lesssim \beta\lambda + \lambda^{-1}\varepsilon_{\text{app}}(\mathcal{F}, \lambda) + \lambda^{-1}\varepsilon_{\text{gen}}(\mathcal{F}, n) + \lambda^{-1}\varepsilon_{\text{gen}}^{\text{reg}}(\nabla\mathcal{F}, m),$$

where $\beta = \|\Delta f_0\|_{L^2(\Omega)}^2 + B_1^2$. The approximation error $\varepsilon_{\text{app}}(\mathcal{F}, \lambda)$, the generalization errors $\varepsilon_{\text{gen}}(\mathcal{F}, n)$ and $\varepsilon_{\text{gen}}^{\text{reg}}(\nabla\mathcal{F}, m)$ are defined, respectively, as

$$\varepsilon_{\text{app}}(\mathcal{F}, \lambda) = \inf_{f \in \mathcal{F}} \left\{ \|f - f_0\|_{L^2(\Omega)}^2 + \lambda \|\nabla(f - f_0)\|_{L^2(\Omega)}^2 \right\},$$

$$\varepsilon_{\text{gen}}(\mathcal{F}, n) = (B_0^2 + \sigma^2)(\log n) \inf_{\delta > 0} \left\{ \left(\frac{2 \log N(B_0\delta, \mathcal{F}, L^2(\mathcal{D}))}{n} \right)^{\frac{1}{2}} + \delta \right\},$$

$$\varepsilon_{\text{gen}}^{\text{reg}}(\nabla\mathcal{F}, m) = B_1^2 \inf_{\delta > 0} \left\{ \max_{1 \leq k \leq d} \frac{\log N(B_{1,k}\delta, D_k\mathcal{F}, L^2(\mathcal{S}))}{m} + \delta \right\}.$$

Approximation error

Let $\Omega \subseteq K \subseteq \mathbb{R}^d$ be two bounded domain. Set the hypothesis class as a deep ReQU neural network $\mathcal{F} = \mathcal{N}(L, W, S)$ with $L = \mathcal{O}(\log N)$ and $S = \mathcal{O}(N^d)$. Then for each $\phi \in C^s(K)$ with $s \in \mathbb{N}_{\geq 2}$, there exists $f \in \mathcal{F}$ such that

$$\begin{aligned}\|f - \phi\|_{L^2(\Omega)} &\leq CN^{-s} \|\phi\|_{C^s(K)}, \\ \|\nabla(f - \phi)\|_{L^2(\Omega)} &\leq CN^{-(s-1)} \|\phi\|_{C^s(K)},\end{aligned}$$

where C is a constant independent of N .

Approximation error

Let $\Omega \subseteq K \subseteq \mathbb{R}^d$ be two bounded domain. Set the hypothesis class as a deep ReQU neural network $\mathcal{F} = \mathcal{N}(L, W, S)$ with $L = \mathcal{O}(\log N)$ and $S = \mathcal{O}(N^d)$. Then for each $\phi \in C^s(K)$ with $s \in \mathbb{N}_{\geq 2}$, there exists $f \in \mathcal{F}$ such that

$$\begin{aligned}\|f - \phi\|_{L^2(\Omega)} &\leq CN^{-s} \|\phi\|_{C^s(K)}, \\ \|\nabla(f - \phi)\|_{L^2(\Omega)} &\leq CN^{-(s-1)} \|\phi\|_{C^s(K)},\end{aligned}$$

where C is a constant independent of N .

Generalization error

Suppose the activation function is piecewise-polynomial. Let $\mathcal{D} = \{X_i\}_{i=1}^n$ and $\mathcal{S} = \{Z_j\}_{j=1}^m$. Then

$$\begin{aligned}\log N(\delta, N(L, S, B), L^2(\mathcal{D})) &\lesssim LS \log(S) \log\left(\frac{nB}{\delta}\right), \\ \log N(\delta, D_k N(L, S, B), L^2(\mathcal{S})) &\lesssim L^2 S \log(S) \log\left(\frac{mB}{\delta}\right).\end{aligned}$$

Under **A1** to **A4**. Let $\Omega \subseteq K \subseteq \mathbb{R}^d$ be two bounded domain. Assume that $f_0 \in C^s(K)$ with $s \in \mathbb{N}_{\geq 2}$. Set the hypothesis class as a deep ReQU neural network class $\mathcal{F} = N(L, W, S)$ with $L = \mathcal{O}(\log n)$ and $S = \mathcal{O}(n^{\frac{d}{d+4s}})$. Let $\lambda = \mathcal{O}(n^{-\frac{s}{d+4s}} \log^2 n)$. Then

$$\begin{aligned}\mathbb{E} \left[\|\hat{f}_{n,m}^\lambda - f_0\|_{L^2(\Omega)}^2 \right] &\leq \mathcal{O} \left(n^{-\frac{2s}{d+4s}} \log^4 n \right) + \mathcal{O} \left(n^{\frac{d}{d+4s}} \log^4 n m^{-1} \right), \\ \mathbb{E} \left[\|\nabla(\hat{f}_{n,m}^\lambda - f_0)\|_{L^2(\Omega)}^2 \right] &\leq \mathcal{O} \left(n^{-\frac{s}{d+4s}} \log^2 n \right) + \mathcal{O} \left(n^{\frac{d+s}{d+4s}} \log^2 n m^{-1} \right).\end{aligned}$$

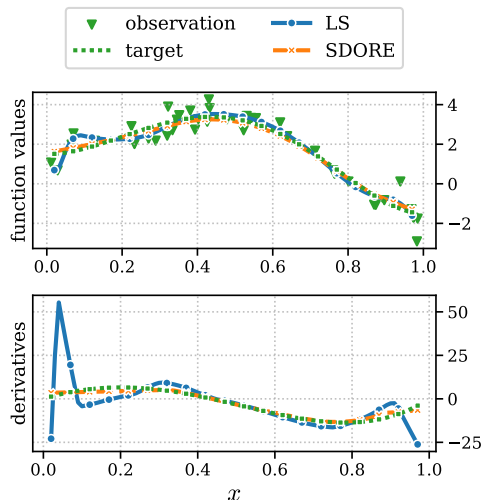
Under **A1** to **A4**. Let $\Omega \subseteq K \subseteq \mathbb{R}^d$ be two bounded domain. Assume that $f_0 \in C^s(K)$ with $s \in \mathbb{N}_{\geq 2}$. Set the hypothesis class as a deep ReQU neural network class $\mathcal{F} = N(L, W, S)$ with $L = \mathcal{O}(\log n)$ and $S = \mathcal{O}(n^{\frac{d}{d+4s}})$. Let $\lambda = \mathcal{O}(n^{-\frac{s}{d+4s}} \log^2 n)$. Then

$$\begin{aligned}\mathbb{E} \left[\|\hat{f}_{n,m}^\lambda - f_0\|_{L^2(\Omega)}^2 \right] &\leq \mathcal{O} \left(n^{-\frac{2s}{d+4s}} \log^4 n \right) + \mathcal{O} \left(n^{\frac{d}{d+4s}} \log^4 n m^{-1} \right), \\ \mathbb{E} \left[\|\nabla(\hat{f}_{n,m}^\lambda - f_0)\|_{L^2(\Omega)}^2 \right] &\leq \mathcal{O} \left(n^{-\frac{s}{d+4s}} \log^2 n \right) + \mathcal{O} \left(n^{\frac{d+s}{d+4s}} \log^2 n m^{-1} \right).\end{aligned}$$

Simultaneously estimate both the regression function f_0 and its gradient ∇f_0 .

- ① Introduction and Background
- ② Sobolev-Penalized Regression
- ③ Semi-Supervised Deep Sobolev Regression
- ④ Applications and Numerical Experiments**
 - Derivative Estimations
 - Nonparametric Variable Selection
 - Inverse Problems
- ⑤ Conclusions

$$f_0(x) = 1 + 36x^2 - 59x^3 + 21x^5 + 0.5 \cos(\pi x)$$



► **A5. Sparsity structure**

There exists $f_0^* : \mathbb{R}^{d^*} \rightarrow \mathbb{R}$ ($1 \leq d^* < d$) such that for each $x := (x_1, \dots, x_d) \in \mathbb{R}^d$,

$$f_0(x_1, \dots, x_d) = f_0^*(x_{j_1}, \dots, x_{j_{d^*}}), \quad \{j_1, \dots, j_{d^*}\} \subseteq [d].$$

► **A5. Sparsity structure**

There exists $f_0^* : \mathbb{R}^{d^*} \rightarrow \mathbb{R}$ ($1 \leq d^* < d$) such that for each $x := (x_1, \dots, x_d) \in \mathbb{R}^d$,

$$f_0(x_1, \dots, x_d) = f_0^*(x_{j_1}, \dots, x_{j_{d^*}}), \quad \{j_1, \dots, j_{d^*}\} \subseteq [d].$$

The derivatives can indicate whether a variable is relevant to the output.

► **A5. Sparsity structure**

There exists $f_0^* : \mathbb{R}^{d^*} \rightarrow \mathbb{R}$ ($1 \leq d^* < d$) such that for each $x := (x_1, \dots, x_d) \in \mathbb{R}^d$,

$$f_0(x_1, \dots, x_d) = f_0^*(x_{j_1}, \dots, x_{j_{d^*}}), \quad \{j_1, \dots, j_{d^*}\} \subseteq [d].$$

The derivatives can indicate whether a variable is relevant to the output.

Relevant variable

- A variable $k \in [d]$ is irrelevant for the function f with respect to Lebesgue measure on Ω , if

$$D_k f(X) = 0 \quad \text{almost surely,}$$

and relevant otherwise. The set of relevant variables is defined as

$$\mathcal{I}(f) = \{k \in [d] : \|D_k f\|_{L^2(\Omega)} > 0\}.$$

	Is f_0 sparse?	Convergence rate
$\mathbb{E}[\ \hat{f}_{n,m}^\lambda - f_0\ _{L^2(\Omega)}^2]$	$\times$	$\tilde{O}(n^{-\frac{2s}{d+4s}})$
$\mathbb{E}[\ \hat{f}_{n,m}^\lambda - f_0\ _{L^2(\Omega)}^2]$	$\checkmark$	$\tilde{O}(n^{-\frac{2s}{d^*+4s}})$
$\mathbb{E}[\ \nabla(\hat{f}_{n,m}^\lambda - f_0)\ _{L^2(\Omega)}^2]$	$\times$	$\tilde{O}(n^{-\frac{s}{d+4s}})$
$\mathbb{E}[\ \nabla(\hat{f}_{n,m}^\lambda - f_0)\ _{L^2(\Omega)}^2]$	$\checkmark$	$\tilde{O}(n^{-\frac{2s}{d^*+4s}})$

	Is f_0 sparse?	Convergence rate
$\mathbb{E}[\ \hat{f}_{n,m}^\lambda - f_0\ _{L^2(\Omega)}^2]$	✗	$\tilde{O}(n^{-\frac{2s}{d+4s}})$
$\mathbb{E}[\ \hat{f}_{n,m}^\lambda - f_0\ _{L^2(\Omega)}^2]$	✓	$\tilde{O}(n^{-\frac{2s}{d^*+4s}})$
$\mathbb{E}[\ \nabla(\hat{f}_{n,m}^\lambda - f_0)\ _{L^2(\Omega)}^2]$	✗	$\tilde{O}(n^{-\frac{s}{d+4s}})$
$\mathbb{E}[\ \nabla(\hat{f}_{n,m}^\lambda - f_0)\ _{L^2(\Omega)}^2]$	✓	$\tilde{O}(n^{-\frac{2s}{d^*+4s}})$

Selection consistency

Under **A1** to **A5**. It follows that

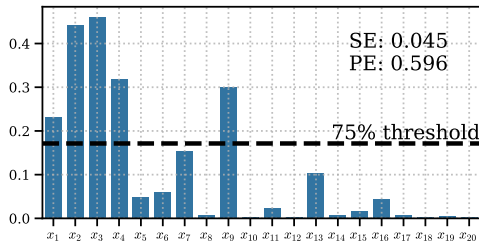
$$\lim_{n \rightarrow \infty} \Pr \left\{ \mathcal{I}(f_0) = \mathcal{I}(\hat{f}_{n,m}^\lambda) \right\} = 1,$$

where $\lambda = \mathcal{O}(n^{-\frac{s}{d^*+4s}} \log^2 n)$, and m is sufficiently large.

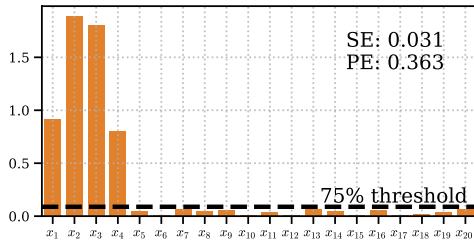
$$f_0(x_1, \dots, x_{20}) = \sum_{i=1}^3 \sum_{j=i+1}^4 x_i x_j.$$

sparse structure

LS



SDORE



Elliptic equation with unknown source

$$\begin{cases} -\nabla \cdot (a(x) \nabla u(x)) + c(x)u = f(x), & \text{in } \Omega, \\ \nabla u \cdot \mathbf{n} = 0, & \text{on } \partial\Omega. \end{cases}$$

Measurement model

$$Y = \mathcal{S}(f^\dagger)(X) + \xi.$$

- ▶ $u^\dagger = \mathcal{S}(f^\dagger)$ is the solution to the elliptic equation.
- ▶ $X \sim \text{unif}(\Omega)$, and $\xi \sim \text{subG}(\sigma^2)$ is the random noise independent of X .

Recovering Procedure

- Sobolev-penalized regression using interior measurements

$$\hat{u}_{n,m}^\lambda \in \arg \min_{u \in \text{conv}(\mathcal{U})} \hat{L}_{n,m}^\lambda(u) = \frac{1}{n} \sum_{i=1}^n (Y_i - u(X_i))^2 + \frac{\lambda}{m} \sum_{j=1}^m \|\nabla u(Z_j)\|_2^2.$$

- Interior position-measurement pairs: $\{(X_i, Y_i)\}_{i=1}^n$. expensive
- Positions variables $Z_1, \dots, Z_m \sim \text{i.i.d. unif}(\Omega)$ very cheap

Recovering Procedure

- Sobolev-penalized regression using interior measurements

$$\hat{u}_{n,m}^\lambda \in \arg \min_{u \in \text{conv}(\mathcal{U})} \hat{L}_{n,m}^\lambda(u) = \frac{1}{n} \sum_{i=1}^n (Y_i - u(X_i))^2 + \frac{\lambda}{m} \sum_{j=1}^m \|\nabla u(Z_j)\|_2^2.$$

- Interior position-measurement pairs: $\{(X_i, Y_i)\}_{i=1}^n$. expensive
- Positions variables $Z_1, \dots, Z_m \sim \text{i.i.d. unif}(\Omega)$ very cheap
- Recovering unknown source using gradient estimator
Find $\hat{f}_{n,m}^\lambda$, such that for each $v \in H^1(\Omega)$,

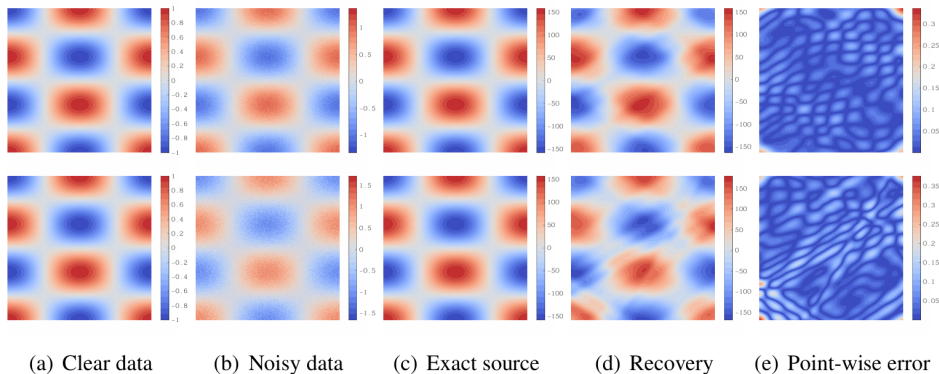
$$(a(x) \nabla \hat{u}_{n,m}^\lambda, \nabla v)_{L^2(\Omega)} + (c(x) \hat{u}_{n,m}^\lambda, v)_{L^2(\Omega)} = (\hat{f}_{n,m}^\lambda, v)_{L^2(\Omega)}.$$

Since $\hat{u}_{n,m}^\lambda \in H^2(\Omega)$, we find

$$\hat{f}_{n,m}^\lambda = -\nabla \cdot (a(x) \nabla \hat{u}_{n,m}^\lambda) + c(x) \hat{u}_{n,m}^\lambda \in L^2(\Omega).$$

in weak norm

$$\mathbb{E} \left[\|\hat{f}_{n,m}^\lambda - f^\dagger\|_{(H^1(\Omega))^*} \right] \lesssim \mathcal{O} \left(n^{-\frac{s}{2(d+4s)}} \right).$$



- ① Introduction and Background
- ② Sobolev-Penalized Regression
- ③ Semi-Supervised Deep Sobolev Regression
- ④ Applications and Numerical Experiments
 - Derivative Estimations
 - Nonparametric Variable Selection
 - Inverse Problems
- ⑤ Conclusions

- ▶ Simultaneously estimations of both the regression function and its gradient.
- ▶ Nonasymptotic convergence rate.
- ▶ Applications in nonparametric variable selection and inverse problems.

Reference: Zhao Ding, Chenguang Duan, Yuling Jiao, and Jerry Zhijian Yang. Semi-Supervised Deep Sobolev Regression: Estimation and Variable Selection by ReQU Neural Network. *IEEE Transactions on Information Theory*, 2025.

Thanks for your attention!

cgduan.math@gmail.com, cgduan.math@whu.edu.cn